

Savitribai Phule Pune University

(Formerly University of Pune)

Bachelors Degree in Data Science

(Faculty of Science and Technology)



Syllabi for B.Sc. (Data Science)

(For Colleges Affiliated to Savitribai Phule Pune University)

Choice Based Credit System (CBCS) Syllabus
Under National Education Policy (NEP)

To be implemented from Academic Year 2024-2025

Savitribai Phule Pune University

Syllabus Structure as per NEP Guidelines

B.Sc. (Data Science) from 2024-25

FY (Level 4.5) SEMESTER II

Course Type	Course code	Course Name	Credits		Teaching Scheme Hrs/Week		Examination Scheme and Marks		
			T H	P R	TH	PR	C E	E E	Total
Subject-1	DS151T	Advanced Python Programming	2	-	2	-	15	35	50
	DS152P	Lab Course on DS151T (Advanced Python Programming)	-	2	-	4	15	35	50
Subject-2	DS153T	Discrete Probability and Probability Distributions	2	-	2	-	15	35	50
	DS154P	Lab Course on DS153T (Discrete Probability and Probability Distributions)	-	2	-	4	15	35	50
Subject-3	DS155T	Graph Theory	2	-	2	-	15	35	50
	DS156P	Lab Course on DS155T (Graph Theory)	-	2	-	4	15	35	50
GE/OE	OE151DSP	Office Automation/ Computer Fundamentals/ Introduction to Google Tools	-	2	-	4	15	35	50
SEC	SEC151DS	Lab Course on Excel and Advanced Excel	-	2	-	4	15	35	50
AEC	AEC101 MAR/HIN/ ENG	MIL-I (Hindi) / MIL-I (Marathi)/ MIL-I (English)	2	-	2	-	15	35	50
VEC	VEC101 ENV	EVS-II	2	-	2	-	50	-	50
CC	CC151PE/ NSS/ NCC	From University Basket	2	-	2	-	50	-	50
Total			12	10	12	20			550

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B.Sc. Data Science (Pattern 2024)
Semester-II

DS151T : Advanced Python Programming

No. of Credits: 2	Teaching Scheme Theory:2 Hrs /Week	Examination Scheme Continuous Evaluation:15 Marks End Semester:35 Marks	
Prerequisites - Fundamentals of Python Programming Language. - Prior knowledge of computational mathematics.			
Objectives - To learn reading, writing and manipulating files - To implement libraries like Pandas, NumPy,SciPy, Matplotlib, Scikit-learn etc. in Python. - To implement the concepts of GUI controls and designing GUI applications. - To learn and know the concepts of file handling, exception handling.			
Course Outcomes On Completion of this course, student will be able to - CO1: Learn reading and writing into files using Python CO2: Design and implement a program to solve a computational problem. CO3: Understand implementation of libraries like Pandas, NumPy, SciPy, Matplotlib, Scikit-learn etc. in Python. CO4: How to handle exceptions and files. CO5: Design and implement GUI application			
Unit No.	Name of Unit	Teaching Hours	CO Targeted
1	File Handling	3	CO1
1.1 Introduction to Files 1.2 Types of Files 1.3 Opening and Closing a Text File 1.4 Writing to a Text File 1.5 Reading from a Text File 1.6 Setting Offsets in a File 1.7 Creating and Traversing a Text File			
2	Python Libraries	17	CO2 & CO3
2.1 Introduction to Python Libraries 2.1.1 Statistical Analysis- NumPy, SciPy, Pandas, StatsModels 2.1.2 Data Visualization- Matplotlib, Seaborn, Plotly 2.1.3 Data Modelling and Machine Learning- Scikit-learn, XGBoost, Eli5 2.1.4 Deep Learning- TensorFlow, Pytorch, Keras 2.1.5 Natural Language Processing (NLP)- NLTK, SpaCy, Gensim			

2.2 Working with Tabular Numeric Data(Numpy with Python) 2.2.1 NumPy Arrays Creation Using array() Function 2.2.2 Array Attributes, NumPy Arrays Creation with Initial Placeholder Content 2.2.3 Integer Indexing, Array Indexing, Boolean ArrayIndexing, Slicing and Iterating in Arrays Basic Arithmetic Operations on NumPy Arrays 2.2.4 Mathematical Functions in NumPy 2.2.5 Changing the Shape of an Array, Stacking and Splitting of Arrays, Broadcasting in Arrays. 2.3 Working with Data Series and Frames 2.3.1 Pandas Data Structures, Reshaping Data, Handling Missing Data 2.3.2 Combining Data, Ordering and Describing Data, Transforming Data, Taming Pandas File I/O 2.4 Plotting Basic Plotting with PyPlot, Matplotlib, Getting to Know Other Plot Types, Plotting with Pandas			
3	Exception Handling	5	CO4
3.1 Python Exception 3.2 Common Exception 3.3 Exception handling in Python (try-except-else) 3.4 The except statement with no exception 3.5 Multiple Exception 3.6 The try-finally clause 3.7 Custom Exception and assert statement			
4	GUI Programming	5	CO5
4.1 Introduction 4.2 Tkinter programming 4.3 Tkinter widgets 4.5 Frame 4.6 Button 4.7 Label 4.8 Entry			
Reference Books			
1.Mark Lutz, Programming Python, O`Reilly, 4th Edition, 2010 2.Dive into Python, Mike 3. Learning Python, 4th Edition by Mark Lutz 4. Programming Python, 4th Edition by Mark Lutz 5.Python Programming:An introduction to computer,John Zelle,3rd Edition. 6. Data Science Essentials in Python: Collect, Organize, Explore, Predict, Value. Dmitry Zinoriev, The Pragmatic Programmers LLC, 2016 7. Introduction to Python Programming. Gowrishankar S., Veena A. CRC Press, Taylor & Francis Group, 2019			

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Semester-II

DS152P : Lab Course on DS151T (Advanced Python Programming)

No. of Credits: 2	Teaching Scheme Practical: 4 Hrs/Week	Examination Scheme Continuous Evaluation: 15 Marks End Semester: 35 Marks	
Prerequisites • Practical Knowledge of Python Programming. • Prior knowledge of computational mathematics.			
Objectives • To learn reading, writing and manipulating files • To implement libraries like Pandas, NumPy, SciPy, Matplotlib, Scikit-learn etc. in Python. • To implement the concepts of GUI controls and designing GUI applications. • To learn and know the concepts of file handling, exception handling.			
Course Outcomes On Completion of this course, student will be able to - CO1: Learn reading and writing into files using Python CO2: Design and implement a program to solve a computational problem. CO3: Understand implementation of libraries like Pandas, NumPy, SciPy, Matplotlib, Scikit-learn etc. in Python. CO4: How to handle exceptions and files. CO5: Design and implement GUI application			
Unit No.	Name of Unit	Hours	CO Targeted
1	File Handling	8	CO1
a. Assignments on reading and writing files b. Assignments on accessing and manipulating files			
2	Python Numpy	12	CO2
a. Assignments on Numpy basics. b. Assignments on Numpy Arrays. c. Assignments on Numpy Linear Algebra d. Assignments on Numpy Statistics e. Assignments on Numpy Strings			
3	Python Pandas	8	CO3
a. Assignments on reading and writing text, CSV files using Pandas			
4	Plotting in Python	12	CO3
a. Assignments on plotting graphs in python using Matplotlib and Pyplot			
5	Assignments on Exception Handling	8	CO4
6	Assignments on GUI	12	CO5

Reference Books

1. Mark Lutz, Programming Python, O`Reilly, 4th Edition, 2010
2. Dive into Python, Mike
3. Learning Python, 4th Edition by Mark Lutz
4. Programming Python, 4th Edition by Mark Lutz
5. Python Programming: An introduction to computer, John Zelle, 3rd Edition.
6. Data Science Essentials in Python: Collect, Organize, Explore, Predict, Value. Dmitry Zinoriev, The Pragmatic Programmers LLC, 2016
7. Introduction to Python Programming. Gowrishankar S., Veena A. CRC Press, Taylor & Francis Group, 2019

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B.Sc. Data Science (Pattern 2024)

DS153T : Discrete Probability and Probability Distributions

No. of Credits: 02	Teaching Scheme Theory: 2 Hours/Week	Examination Scheme Continuous Evaluation:15 Marks End Semester : 35 Marks	
Prerequisites Basics of Set theory, Mathematical operations			
Objectives - To revise the basic concepts of probability, axiomatic theory of probability. - To understand the concept of random variable - To study probability distribution (univariate and bivariate) discrete random variables, expectation and moments of probability distribution - To find marginal distribution and conditional distribution of bivariate frequency distribution - To find conditional mean of bivariate frequency distribution - To find variance, covariance and correlation of bivariate frequency distribution			
Course Outcomes On Completion of this course, student will be able to – CO1: Find the probabilities of events and its expectation, mean, variance, etc. CO2:Distinguish between random and non-random experiments CO3:Identify the nature of distribution CO4: Find marginal distribution and conditional distribution CO5: Find mean of marginal distribution and conditional mean of bivariate frequency distribution CO6: Find correlation of bivariate frequency distribution			
Unit No.	Name of Unit	Teaching Hours	CO Targeted
1	Introduction to Probability	07	CO1, CO2
Basics of Probability: Experiments/Models, Ideas of deterministic and non-deterministic models. Random Experiment.			
Definitions: Sample space, Discrete sample space: finite and countably infinite, Event, Elementary event, Complement of an event, Certain event, Impossible event.			
Occurrence of events: Concept of occurrence of an event. Algebra of events and its representation in set theory notation. Occurrence of following events (i) at least one of the given events, (ii) none of the given events, (iii) all of the given events, (iv) mutually exclusive events, (v) mutually exhaustive events, (vi) exactly one event out of the given events.			
Classical definition of probability and its limitations. Probability model, probability of an event, equiprobable and non-equiprobable sample space.			
Axiomatic definition of probability. Theorems and results on probability with proofs based on axiomatic definition such as $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Generalization $P(A \cup B \cup C), 0 \leq P(A) \leq 1, P(A) + P(A') = 1, P(\Phi) = 0$ and when $A \subseteq B$ then $P(A) \leq$			

$P(B)$.			
2	Conditional Probability	05	CO2
Definition of conditional probability of an event. Results on conditional probability. Definition of independence of two events $P(A \cap B) = P(A)P(B)$. Pairwise independence and mutual independence for three events. Multiplication theorem $P(A \cap B) = P(A)P(B A)$. Generalization to $P(A \cap B \cap C)$. Partition of the sample space, prior and posterior probabilities. Proof of Bayes' theorem. Applications of Bayes' theorem in real life.			
3	Univariate Probability Distributions and its Mathematical Expectation	9	CO3
Univariate Probability Distributions defined on Discrete Sample Space: Concept and definition of a discrete random variable. Probability Mass Function (<i>pmf</i>) and cumulative Distribution Function (<i>cdf</i>), $F(\cdot)$ of discrete random variable, properties of <i>cdf</i>. Mode and median of a univariate discrete probability distribution. Mathematical Expectation: Definition of expectation (mean) of a random variable, expectation of a function of a random variable, Moment Generating Function (<i>mgf</i>) and Cumulative Generating Function (<i>cgf</i>). Properties of <i>mgf</i> and <i>cgf</i>. Definitions of variance, standard deviation (SD) and Coefficient of variation (CV) of univariate probability distribution, effect of change of origin and scale on mean, variance and SD. Definition of raw, central and factorial raw moments of univariate probability Distributions and their interrelations (without proof). Coefficients of skewness and kurtosis based on moments.			
4	Mathematical Expectation for Bivariate Frequency Distribution	9	CO4, CO5, CO6
Definition of raw and central moments, <i>mgf</i> and <i>cgf</i> . Theorems on expectations of sum and product of two jointly distributed random variables. Conditional expectation. Definitions of conditional mean and conditional variance. Definition of covariance, coefficient of correlation, independence and uncorrelatedness of two variables. Variance of linear combination of variables, $Var(aX + bY)$, $Var(aX + bY + C)$ and its generalization.			
Reference Books			
1. Gupta, S. C. and Kapoor, V. K. (1983). Fundamentals of Mathematical Statistics, Eighth Edition, Sultan Chand and Sons Publishers, New Delhi. 2. Sarma, K. V. S. (2001). Statistics Made it Simple: Do it yourself on PC. Prentice Hall of India, New Delhi. 3. Hoel, P. G. (1971). Introduction to Mathematical Statistics, John Wiley and Sons, New York. 4. Hogg, R. V. and Craig, R. G. (1989). Introduction to Mathematical Statistics, Ed. MacMillan Publishing Co., New York. 5. Mayer, P. (1972). Introductory Probability and Statistical Applications, Addison Wesley Publishing Co., London. 6. Mood, A. M. and Graybill, F. A. and Boes D. C. (1974). Introduction to the Theory of Statistics, Ed. 3, McGraw Hill Book Company. 7. Rao, VLS Prakash (2008). First Course in Probability and Statistics, New Age International Publishers, New Delhi. 8. Ross S. (2002). A First Course in Probability, Sixth Edition, Pearson Education, Inc. & Dorling Kindersley Publishing, Inc.			

SavitribaiPhule Pune University
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Semester-II

DS154P : Lab Course on DS153T (Discrete Probability and Probability Distributions)

No. of Credits: 02	Teaching Scheme Practical: 4 Hours/Week	Examination Scheme Continuous Evaluation:15 Marks End Semester : 35 Marks
Prerequisites Basics of Set theory, Mathematical operations		
Objectives - To understand the concept of random variable - To study probability distribution (univariate and bivariate) discrete random variables, expectation and moments of probability distribution - To find marginal distribution and conditional distribution of bivariate frequency distribution - To find conditional mean of bivariate frequency distribution - To find variance, covariance and correlation of bivariate frequency distribution		
Course Outcomes On Completion of this course, student will be able to – CO1: Find the probabilities of events and its expectation, mean, variance, etc. CO2:Distinguish between random and non-random experiments CO3:Identify the nature of distribution CO4: Find marginal distribution and conditional distribution CO5: Find mean of marginal distribution and conditional mean of bivariate frequency distribution CO6: Find correlation of bivariate frequency distribution		
Sr.No.	List of Practical Assignments	Hours
1	Calculation of probability for different events based on real life situations.	4
2	Calculation of mathematical expectation and variance.	4
3	Obtain marginal and conditional distribution of bivariate probability distribution	4
4	Calculation of conditional expectation and conditional variance	4
5	Calculation of variance of linear combination.	4
6	Checking the independence of the probabilities of the events	4
7	Calculation of the correlation coefficient based on bivariate probability distribution	8
8	Model sampling from the given probability distributions	12
9	Small Project equivalent to 4 practical.	16

Reference Books

1. Gupta, S. C. and Kapoor, V. K. (1983). Fundamentals of Mathematical Statistics, Eighth Edition, Sultan Chand and Sons Publishers, New Delhi.
2. Sarma, K. V. S. (2001). Statistics Made it Simple: Do it yourself on PC. Prentce Hall of India, New Delhi.
3. Hoel, P. G. (1971). Introduction to Mathematical Statistics, John Wiley and Sons, New York.
4. Hogg, R. V. and Craig, R. G. (1989). Introduction to Mathematical Statistics, Ed. MacMillan Publishing Co., New York.
5. Mayer, P. (1972). Introductory Probability and Statistical Applications, Addison Wesley Publishing Co., London.
6. Mood, A. M. and Graybill, F. A. and Boes D. C. (1974). Introduction to the Theory of Statistics, Ed. 3, McGraw Hill Book Company.
7. Rao, VLS Prakash (2008). First Course in Probability and Statistics, New Age International Publishers, New Delhi.
8. Ross S. (2002). A First Course in Probability, Sixth Edition, Pearson Education, Inc. & Dorling Kindersley Publishing, Inc.

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Semester-II

DS155T : Graph Theory

No. of Credits: 2	Teaching Scheme Theory: 2 Hrs/Week	Examination Scheme Continuous Evaluation: 15 Marks End Semester : 35 Marks	
Prerequisites Basics of mathematics, Set Theory			
Objectives - To introduce students about graph, graph models, types of graph, connectivity, applications of graph theory. - To know how to find shortest path for different Eulerian and Hamiltonian circuit. - To introduce students about Trees, applications of trees, binary tree, tree traversal , spanning trees. - To know how to find minimum spanning trees. - To make students familiar with the use of all these concepts as tools in other areas of the course curriculum.			
Course Outcomes On Completion of this course, student will be able to : CO1: Understand the graph, and graph models, terminology of graph. CO2: Students can solve examples on adjacency and incidence matrix. CO3: Identify the Euler tours and Hamiltonian cycle and find shortest path. CO4: Able to Compute the shortest spanning trees. CO5: Students can solve the problems on tournaments and traffic flow.			
Unit No.	Name of Unit	Teaching Hours	CO Targeted
1	Graphs and Graph Models	4	CO1, CO5
Graph: Definition, basic terminology of Graph, Graph Models, Social networks, Communication networks, Information networks ,Software Design Applications, Transportation networks, Biological networks, Tournaments.			
2	Graph Isomorphism	5	CO2
Handshaking lemma, Special Types of Graph, Directed graph, Matrix representation of graph, Definition of isomorphism,Examples on isomorphism of graphs.			
3	Connected Graph	8	CO3
Walk, trail, path, cycle, connected graph, disconnected graph, component, Cut edge, Cut vertex, Cut set, Vertex connectivity, edge connectivity, Minimal degree of a graph, Relation between Vertex connectivity, edge connectivity and Minimal degree of a graph,Weighted graph, Shortest path algorithm, Dijkstra's algorithm			
4	Eulerian and Hamiltonian Graphs .	5	CO3
The Konigsberg Seven Bridge problem, Euler’s path, Euler’s circuit, Eulerian graph, Fleury’s algorithm, Hamilton path, Hamilton Circuit, Hamiltonian graph, Applications of Eulerian and			

Hamiltonian graph: Traveling Salesman problem, Chinese Postman problem.			
5	Trees	8	CO4
Definition of tree, basic terminology of tree, properties of trees, Eccentricity of a vertex, Centre, diameter, radius of a tree, Spanning Tree, Chords and branches of Spanning Tree, Shortest spanning tree, Kruskal's algorithm, M-ary tree, binary tree, Tree traversal, Ordered rooted tree, polish notation, arborescence.			
Reference Books			
1. Kenneth Rosen, Discrete Mathematics and It's Applications, Tata McGraw Hill, Seventh Edition. 2. Narsingh Deo, Graph Theory with applications to computer science and engineering, Prentice Hall. 3. Dougals B. West, Introduction to Graph Theory, Pearson Education, Second edition.			

Savitribai Phule Pune University
B.Sc. Data Science (Pattern 2024)
Semester-II
DS156P : Lab Course on DS155T (Graph Theory)

No. of Credits: 2	Teaching Scheme Practical: 4 Hours/Week	Examination Scheme Continuous Evaluation: 15 Marks End Semester : 35 Marks
Prerequisites Basics of mathematics, Set Theory		
Objectives - To introduce students about graph, graph models, types of graph, connectivity, applications of graph theory. - To know how to find shortest path for different Eulerian and Hamiltonian circuit. - To introduce students about Trees, applications of trees, binary tree, tree traversal, spanning trees. - To know how to find minimum spanning trees. - To make students familiar with the use of all these concepts as tools in other areas of the course curriculum. - To know how to use Maxima software.		
Course Outcomes On Completion of this course, student will be able to : CO1: Understand the graph, and graph models, terminology of graph. CO2: Students can solve examples on adjacency and incidence matrix. CO3: Identify the Euler tours and Hamiltonian cycle and find shortest path. CO4: Able to Compute the shortest spanning trees. CO5: Students can solve the problems on tournaments and traffic flow. CO6: Students can solve the problems on theory using Maxima Software.		
Sr.No.	List of Practical Assignments	Hours
1	Problem Solving on Unit 1: Graphs and Graph Models (Written)	4
2	Problem Solving on Unit 2: Graph Isomorphism (Written)	4
3	Problem Solving on Unit 3: Connected Graph (Written)	4
4	Problem Solving on Unit 4: Eulerian and Hamiltonian Graphs (Written)	4
5	Problem Solving on Unit 5: Trees (Written)	4
6	Problem Solving on Unit 1: Graphs and Graph Models (Using Maxima Software)	8
7	Problem Solving on Unit 2: Graph Isomorphism (Using Maxima Software)	8
8	Problem Solving on Unit 3: Connected Graph (Using Maxima Software)	8
9	Problem Solving on Unit 4: Eulerian and Hamiltonian Graphs (Using Maxima Software)	8
10	Problem Solving on Unit 5: Trees (Using Maxima Software)	8

Reference Books
1. Kenneth Rosen, Discrete Mathematics and It's Applications, Tata McGraw Hill, Seventh Edition. 2. Narsingh Deo, Graph Theory with applications to computer science and engineering, Prentice Hall. 3. Dougals B. West, Introduction to Graph Theory, Pearson Education, Second edition.

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Semester-II

SEC151DS : Lab Course on Excel and Advanced Excel

No. of Credits: 02	Teaching Scheme Practical: 4 Hrs/Week	Examination Scheme Continuous Evaluation:15 Marks End Semester : 35 Marks	
Prerequisites Basic Computer Skills and Mathematics Skill.			
Objectives - To familiarize the student in introducing and exploring MS excel. - To provide different ways of representation and exploratory data analysis in excel. - To prepare the students to use excel in their project works - Analyze data like a professional.			
Course Outcomes On Completion of this course, student will be able to - CO1: To Implement fundamental concept of Microsoft Excel CO2: Perform calculations in excel and apply excel functions. CO3: Represent data using charts and diagrams CO4: Design advanced graphic presentations on stored data. CO5: Perform various advanced data tools and data analytics.			
Unit No.	Name of Unit	Teaching Hours	CO Targeted
1	Introduction to Microsoft Excel	7	CO1
- Concepts of Work book &Work sheets - Various Data Types - Using different features with data, Cell and Texts - Inserting, Removing & Resizing of Columns & Rows - Working with Data and Ranges - Entering data into worksheet - Saving & quitting worksheet - Opening and moving around in an existing worksheet - Toolbars and menu, keyboard shortcuts - Working with single and multiple workbook- copying, renaming, moving, adding and deleting, copy in gentries and moving between work books - Different Views of Work sheets - Column Freezing, Labels, Hiding, Splitting etc. - Using different features with Data and Text; Advanced paste special techniques			
2	Formulas ,Functions and charts in Excel	7	CO1,CO2,CO3
Use of Formulas			

• Calculations and Functions • Chart Tools • Different types of charts and their use • Logical Functions • Text Functions • Date and Time Functions • Lookups.			
3	Advance Data Tools	4	CO5
• What-if-Analysis- Goal Seek, Data Table • Scenario Manager • Formatting Charts, 3D Graphs			
4	Advanced Graphing and Charting	5	CO3,CO4
• Formatting and customizing Pivot tables • Using advanced options of Pivot tables, Pivot charts • Line, Bar and Pie charts • Scatter plots • Histograms.			
5	Analytics using Excel	7	CO5
• Data analysis using normal chart • Regression in Excel • Correlation, stddev, average, ANOVA			
Reference Books			
1. Mastering MS Excel: Functions and Formulas, Webtech (Khanna Publications) 2. Microsoft Excel 2019 Data Analysis and Business Modeling, Wayne Winston, 2019 3. Advance Excel 2016, training Guide, By Ritu Arora			